

SVGA Projectors			
Panasonic	Sharp	Benq	Zen
PT-LB60EA	XG-C330X	MP611c	PTV-01B
17.26	17.12	10.11	10.85
LCD	LCD	DLP	LCD
1024x768	1024x768	800x600	800x600
3200/400:1	3300/600:1	2100/2000:1	1200/500:1
4:3/16:9	4:3/16:9	4:3/16:9	4:3/16:9
X/1.2:1	X/1.2:1	X/1.15:1	X/NA
33-300	40-500	31-300	38-150
1.1-10.8	0.56-8.44	NA	1.2-3.8
2000/3000	2000/3000	3000/4000	4500/6000
✓	✓	X	X
15-91	15-70	31-82	NA
50-85	43-85	48-85	NA
✓/✓/✓/✓	✓/✓/✓/✓	✓/✓/✓/✓	✓/✓/✓/✓
✓/✓	✓/✓	✓/✓	X/✓
X	✓	✓	X
✓	X	X	X
2.6	4.25	2.67	5.80
220/300	275 / 375	200/285	150/225
35	36	28	40.00
7.5	8	6.5	6.50
✓/✓	✓/✓	✓/✓	X/✓
50.35	50.24	42.23	26.81
8/8	7.5/7.75	6.75/7	4.75/4.5
8.25	8.25	7.25	4.00
8/8.5	8/8.25	8/7.75	5.25
8.50	8.25	8.00	5.50
8.50	8.25	7.75	5.50
8.00	7.50	6.50	5.50
8.50	8.00	7.50	5.00
8.50	8.75	8.00	4.50
8.25	8.00	4.50	4.25
7.5/5	8.25/8	5.75/6	4.00
7.5/8/8.25	8/7.75/8.25	6.75/6.5/7.25	5/4.5/4.5
7.75/6	7.75/8	7/7.5	4.75/6
6.00	8.00	7.50	4.50
8.00	8.00	5.25	4.75
7.75	8.00	7.50	6.00
8.5/8	7.5/7.75	5.33/5	5/4.75
8.5/8.25	8/8.25	6/6.5	4.75/3.75
8.25	8.25	7.00	5.50
8.00	8.00	6.25	3.25
7.25/8	6.75/7.75	6.5/6	3.5/3
9.00	8.75	8.75	5.25
164,990	137,000	52,000	42,000
25,000	19,500	18,000	3,000
4.55	5.47	6.73	8.33
72.17	72.83	59.08	45.99
2	2	2	2



+ Strong overall performance
- Slightly bulky



+ Good text rendition
- Poor contrast ratio



+ Low price
- Poor performance



+ Very low price
- Extremely poor performance

Jargon Buster

Resolution Unlike CRT monitors and most TFT displays today, projectors are finicky about native resolutions and will not perform well at anything less (or more) than their native values. Quite simply, the higher the resolution the better the performance—smaller and more precise pixels lead to greater image quality with less aliasing. An XGA projector will, 9.9 times out of 10, outperform an SVGA projector. However, value also makes an important point here, and the price difference between the projectors sporting these resolutions is very large.

Contrast Ratio Quite simply, the higher the contrast ratio the better. Contrast ratio is defined as the ratio between the brightest and darkest areas of a screen. It represents the maximum variations in the intensity of light that a projector can render. A contrast ratio of 400:1 is sufficient for most applications. Movie buffs and gamers should look at something in the range of 600:1.

Keystone Correction Keystone occurs when the distance between the projector and the top of the image on-screen is much greater than the distance between the projector and the bottom of the screen. Keystone is also called Trapezoidal Distortion. Keystone correction, as the name suggests, consists of correcting the distortion by altering the shape of the projected image. The Keystone correction value is measured in degrees positive or negative.

Moiré Moiré is a phenomenon noticed when two screened images are superimposed on each other at certain angles. In terms of projectors and displays in general, moiré will cause one pattern to be superimposed on another, making for somewhat of a blurred effect. It is an undesirable effect and is present in most projectors. Only the degree of severity differs, based on how visible (or not) it is.

Video Interconnects This is significant especially for home entertainment users. D-Sub and DVI (Digital Visual Connectivity) are mainly useful for PC interfaces. The difference is that DVI provides a digital signal, hence offers higher quality viewing especially for multimedia content. As far as connectivity to other devices like DVD players go S-Video and Composite connects are something to look for. They offer much better colour separation than RCA. HDMI (High Definition Media Interface) is an emerging standard that carries uncompressed digital video and audio signals on a single connector.

Focus Matrix DisplayMate has a whole series of focus tests under its Sharpness and Resolution test suite to test the ability of a projector or screen to display sharp, finely-focused images. These test screens consist of fine patterns of dots in differing dot densities, and also minute, identical patterns. (There are different screens, some with finer patterns, some with coarser patterns.) An ideal projector will render these minute details with as little variation as possible on a single screen. We check for variance in patterns on a single screen.

Aliasing Definitions abound, but simply put, aliasing in terms of digital imaging refers to the occurrence of jagged edges that occur when a signal is sampled and reconstructed as an alias of another signal (the original). Its also referred to as artefacting.